

UNIVERSITY OF TWENTE

MECHANICAL ENGINEERING

MODULE 10 - AIRCRAFT ENGINEERING

Conceptual design of an unmanned aerial vehicle for disaster relief

Abstract

This paper presents a conceptual design of a fixed wing, unmanned aerial vehicle for disaster relief. It is able to transport payloads of 25 kg and deliver it to a location up to 50 km away. This paper explores the aerodynamics behaviour and flight mechanics of the aircraft by analysing the performance of a NACA 4412 airfoil in Xfoil and its implementation in the design. The main features of the aircraft is a rectangular fuselage that converges to a two rail system upon which an empennage with stabilizers modelled as NACA 0012 airfoils is mounted. The aircraft has a total drag of 41.1 N during cruise conditions, and a $(C_L/C_D)_{max} = 14.6$ at an angle of attack of 3.7° . One tractor propeller driven by a 20 kW motor located at the nose of the aircraft provides thrust to cruise and climb with a speed of 42 m/s and 36 m/s respectively. In the report, the structure of the aircraft, consisting of Al 7075 and Al 2024, details the feasibility and structural limits of the design. The design report is summarized with a business case for potential investors and customers.

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1 Introduction

Emergency response organizations face difficulties in delivering small aid packages to individuals in remote and hard to reach areas, leaving air transport as the only life-saving option. This report documents a conceptual design of an unmanned aircraft for *Twente In The Air Netherlands* (TITAN), that aim to address this issue. The design has been tailor-made for transporting a payload of 25 kg up to 50 km away, making it the ideal solution for such situations.

Certain specification have been prioritized during the design process to optimize the aircraft for its application. These requirements include; an optimal cruising velocity of 150 km/h for fast delivery times and a minimum velocity of 100 km/h to allow for accurate package drops. The aircraft should have easily accessible payload and battery compartment, resulting in a turn around time of no longer than 10 minutes. Furthermore, a high rate of climb of at least 3.0 m/s is desired to manoeuvre the plane in mountains terrain. Most importantly, the aircraft's design is to emphasize modularity and low cost, making it appealing to potential customers. Along with these specifications, the airplane is to meet all the requirements set by TITAN, that can be found in the project description, [1].

In the upcoming report, the design is introduced through an initial overview of the aircraft and weight estimation, before diving into the performance section that divulges the flight mechanics and aerodynamic behaviour of the design. Structural calculations are shown next, to demonstrate the feasibility of the design, along with a CAD ¹ model used for illustration. Further details are presented in the Appendix, specifically in Section 9, where a comprehensive business case is outlined to underscore the economic viability and strategic considerations associated with the proposed aircraft design.

2 Overview of Aircraft

Before diving into detailed calculations, some general decisions regarding the layout and configuration have been made for this application. These decisions are presented in this section.

¹Computer Aided Design

2.1 Flight Profile

The main objective of the airplane is characterized by high-speed travel to the target area, followed by a payload release and return to take-off location. Fig. 1 illustrates this objective using a flight profile.

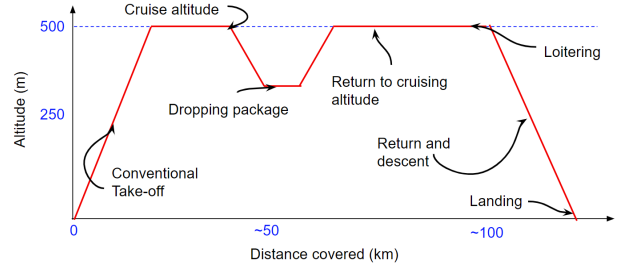


Figure 1: Estimated flight profile.

TITAN and its customer base is located in Europe, where the majority of countries have an average altitude of less than 300 m [2]. For this reason, a desired cruise altitude of 500 m above the sea level has been chosen, as it will allow the airplane to reach a large portion of the European plain. EASA certified airplanes of similar size operating in Europe fly at an altitude from 270 to 900 m [3], which is another reason why this cruise altitude has been chosen.

2.2 Layout Configuration

Layout decisions have been primarily made based on three conditions; the ease of accessing the payload and battery compartment, the stability on the ground and, the cost of production. All of these aim to optimize for the before mentioned specifications.

A high wing configuration has been chosen, so that the payload is easier to load from the back side of the airplane's fuselage. In addition, a high wing configuration sets the center of gravity below the wings, making it less susceptible to rolling [4]. Rectangular wings have also been chosen as the wing configuration due less complex manufacturing, reducing the overall costs of the aircraft.

To further assist with cargo loading, the fuselage is configured so that two long rails connect the main storage compartment with the tail of the airplane. This drastic reduction in cross-section of the fuselage reduces the need for material while making the storage compartment more accessible. It is also a key feature that makes the airplane modular, meaning the tail and rails can be easily

detached from the fuselage and used for compact storage. This feature is explored in more detail in the business case, in Section 9.

The two rail system makes the high boom tail the ideal choice for this fuselage configuration due to the mounting system. It comprises two vertical stabilizers stemming from the ends of the rods, that are connected by one horizontal stabilizer on the top. The downside of this tail is the greater weight from the two vertical stabilizers that relocate the center of gravity further down the plane.

2.3 Propulsion, breaking and control

The power source requirement from TITAN restricts the propulsion system to a propeller-driven aircraft, which is well-suited for low altitudes and low speeds [4], either way. In this scenario, the propellers have been positioned in the tractor configuration. This choice is influenced by the offsetting of the center of gravity towards the front due to the heavy tail. The propeller also benefits from an undisturbed air stream in this configuration.

To control the aircraft, a set of ailerons, two rudders and an elevator are implemented for basic control. The choice of excluding flaps and slats which are not mission critical components for 500 m cruise altitude has been made to reduce costs.

On the other hand, a combination of disk brakes and ailerons will be used as the breaking system, for landing. A tricycle landing gear is chosen, primarily due to the fuselage layout.

2.4 System-Layout

To give an overview of the aircraft's avionics, a system diagram, shown in Fig.2 has been created.

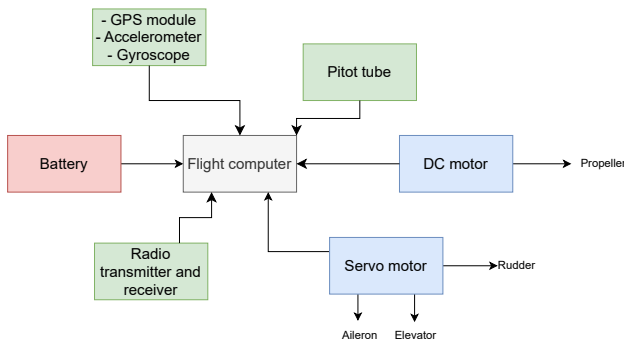


Figure 2: System diagram of aircraft.

The aircraft will have an onboard radio transmitter and receiver, allowing it to receive and send up-to-date flight path information to the operator. This is a necessary inclusion that not only

allows for autonomous flight but also as a safety precaution.

3 Weight estimation

Table 1 shows the first and final weight estimations of the aircraft. According to the initial weight estimation, showed in the second column, the aircraft will have a weight of 150 kg. This weight estimation was performed by estimating the weight of different important components the aircraft. The final weight estimation, showed in the third column, is based on the calculations that have been made and will be explained throughout this report. As can be seen from the table, this final weight estimation is equal to 143 kg.

Table 1: Weight estimation.

Component	Initial weight (kg)	Final weight (kg)
Motor	17	17
Battery	62.5	35.5
Rods	-	6.6
Propeller	-	1.4
Payload	25	25
Wings	18	22
Fuselage	10	13.8
Empennage	4	9.2
Flight electronics	5.5	4.4
Landing gear	8	8.1
Total:	150	143

4 Performance

4.1 Aircraft components

This section explores the performance of the design and justifies design choices through considerations with aerodynamics theory.

4.1.1 Flight parameters

The following flight parameters have been calculated for the desired optimal cruise velocity of 150 km/h ≈ 41.6 m/s, a cruise altitude of 500 m and the initial weight estimation of 150 kg for reasons that have been stated above. Once the new

weight estimation has been obtained, the calculations were remade in the exact same fashion. The *Standard Atmosphere Model* listed in the appendix of [4] has been used for values of air density at different altitudes.

4.1.2 Infinite Wings

A wing's primary objective is to create lift, which is why selecting an adequate airfoil with a high lift coefficient and forgiving stall characteristics is essential. The NACA 4412 has been selected as the airfoil of choice because it is a modern low-speed airfoil that can be analysed in XFOIL [5]. XFOIL is a program that uses the panel method, meaning it divides the preset airfoil into sections and iteratively solves the potential flow equations and the boundary layer equations to find a converged solution. To compute the performance of the airfoil at the cruising altitude using XFOIL, the Reynolds number has been calculated for incompressible flow at a chord length of 0.5 m. To determine the viscosity at a cruising altitude of 500 m, Sutherland's Law [6] has been used in combination with the standard atmosphere model.

$$Re = \frac{\rho_{\infty} V_{\infty} c}{\mu_{\infty}} = 1.370 \times 10^6 \quad (1)$$

A chord length of 0.5 m has been chosen not only for convenience but also to retain a small thickness of the airfoil at 6cm. From XFOIL the aerodynamic coefficients for the NACA 4412 airfoil across a range of angles of attack are shown in Fig. 3.

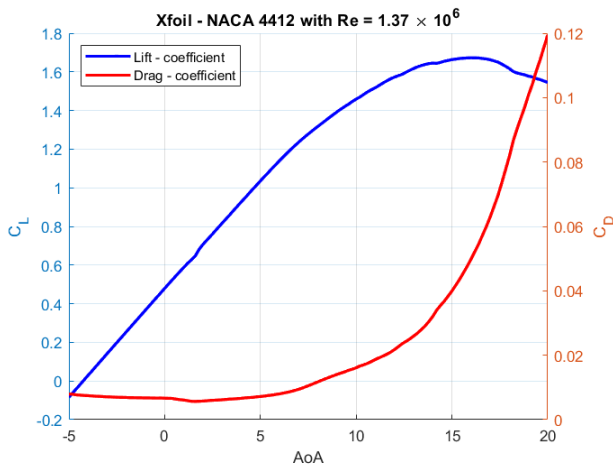


Figure 3: XFOIL - NACA 4412 lift and drag coefficients.

4.1.3 Finite Wings

To determine the other parameters of the wing, the assumption is made that the wing is the only

lift-generating device on the aircraft. Coupled with the force balance for stable unaccelerated flight, the lift produced by the wings must be equal to the weight of the aircraft [4], $L = W$. For an initial calculation, the wing surface area S is assumed to be 2 m^2 and will be optimized for once a more accurate weight estimation has been determined.

To retain low costs of production, less complex to manufacture rectangular shaped wings have been chosen, over better preforming shapes such as the elliptical wings [4]. Rectangular wings create more induced drag, that is reflected by a lower value in the Oswald efficiency e . Calculations of the Oswald efficiency are outside the scope of this report, instead the Oswald efficiency is assumed to be 0.75 as suggested by [7]. More details on this topic are shown in Chapter 6: Three-Dimensional Incompressible Flow[7].

A high aspect ratio wing enables a sustained, endurance flight and a low aspect ratio wing offers high maneuverability [4]. Both advantages are necessary for this type of aircraft and for this reason, a medium aspect ratio AR wing of 8 has been chosen. The wingspan b , for such an aspect ratio and chord length, is equal to 4 m.

With the parameters of the wing known, and the zero-lift drag obtained from XFOIL as shown in Fig. 3, the drag polar is used to calculate the total drag. The assumption that the zero-lift drag coefficient is the same as the minimum drag coefficient is made.

$$C_L = \frac{L}{\frac{1}{2} S \rho_{\infty} V_{\infty}^2} = \frac{W}{\frac{1}{2} S \rho_{\infty} V_{\infty}^2} \quad (2)$$

$$D = \frac{1}{2} \rho_{\infty} V_{\infty}^2 S \left(\underbrace{C_{D,0}}_{\text{Zero-lift drag}} + \underbrace{\left(\frac{C_L^2}{\pi e AR} \right)}_{\text{lift-induced drag}} \right) \quad (3)$$

This gives a drag force of the wing of 16.95 N. Using Eq. 2 and Eq. 3, the process is then reverse to determine the lift coefficient and drag force without the payload, and are found to be 0.58 N and 13.38 N respectively. Using Fig.3, the angle of attack to achieve this lift coefficient is found to be 0.95° .

The design process was continued using the values obtained from the initial calculations, before optimizing for the wing dimensions with the new weight estimations. The optimized wing parameters can be seen in Table. 2.

Table 2: Final optimized wing dimensions.

Chord length c	0.5m
Surface area S	1.96m ²
Wingspan b	3.92m
Aspect ratio AR	7.84

4.1.4 Fuselage

A rectangular fuselage design was chosen, for a more stable and easier to manufacture mounting connection with the wings. Because the aircraft is unmanned, pressurization of the cabin for which a circular fuselage is preferred, is not necessary in this design. Another advantage of a rectangular fuselage is the placement of the payload and the battery inside the fuselage. The final dimensions for the fuselage are: a width of 0.8 m, a height of 0.6 m and a mean length of 1.14 m. The drag of the fuselage will be calculated by calculating the friction drag of the wetted area. This wetted area is equivalent to:

$$S_{wetted} = l \cdot (2 \cdot w + 2 \cdot h) \quad (4)$$

Resulting in a wetted area of 3.19 m². The total friction drag can then be calculated using:

$$D_{f,skin} = C_{f,skin} \frac{1}{2} \rho_{\infty} V_{\infty}^2 S_{wetted} \quad (5)$$

Where C_f for a turbulent flat plate estimation equals:

$$C_{f,skin} = 0.074 \cdot \left(\frac{\mu_{\infty}}{\rho_{\infty} U_{\infty} l} \right)^{1/5} \quad (6)$$

This gives that the skin friction drag of the fuselage is equal to 12.02 N. However, the fuselage does not only experience skin friction drag, but also drag caused by the frontal area of the fuselage. The drag of the frontal area of the fuselage is calculated using:

$$D_{f,frontal} = C_{f,frontal} \frac{1}{2} \rho_{\infty} V_{\infty}^2 A \quad (7)$$

where, A is the frontal cross-sectional area, 0.48 m². Based on the dimensions of the fuselage, it can be assumed to behave as a flat plate in an air stream, where the length is significantly longer than the frontal area. Such a flat plate experiences a drag coefficient of 0.01 according to [7] [8]. This gives a frontal area drag of 4.86 N.

The last part of the fuselage, are the rods. These

connect the empennage to the fuselage of the plane. These also generate skin friction drag. Therefore, Equation 5 and 6 can be re-used for the rods, where the wetted area is now the circumference of one rod, multiplied with the length. The radius of the rods is equal to 4 cm and the length is 1 m giving an area of 0.25 m². With such a configuration, the rods generate a drag of 1.94 N and the total drag of the fuselage is equal to 18.83 N.

4.1.5 Landing gear

The aircraft has three landing gears. One is located in the front of the airplane and the other two are located at the rear side of the fuselage. From calculations, it was found that the bars connecting the fuselage and the landing gear wheel have significantly less drag when they are made in an airfoil shape compared to cylindrical bars. Therefore, it is chosen to make these bars in the shape of an airfoil, specifically, the NACA 0012 airfoil.

The drag of this airfoil is calculated with XFOIL, where it is assumed that the free stream velocity is always hitting the airfoil with an angle of attack of zero degrees. In accordance with the classical thin airfoil theory for symmetrical airfoil, no lift is generated at $\alpha = 0$ and hence no lift-induced drag. However, the lack of induced drag does not mean zero drag as in *d'Alembert's paradox* for infinite wings, because the zero-lift drag is still present [7].

So to calculate the drag of the landing gear bars, the drag coefficient calculated by XFOIL is multiplied with the dynamic pressure and the surface area of the airfoil. The chord of the airfoil is 0.17 m and the span of the airfoil is 0.2 m. The chord of the airfoil is based on the forces acting on the landing, which will be explained later in the report. In the end, the drag of the landing gear bars is calculated to be 0.64 N. The drag of the landing gear wheels are neglected in this concept.

4.1.6 Empennage

The empennage consists of two parts: the horizontal stabilizer and the vertical stabilizer. the symetical NACA 0012 airfoil has been chosen for Both stabilizers to produce zero lift when the angle of attack is zero. For the horizontal stabilizer, the chord is set to 0.35 m and the span is set to 1.5 m. These dimensions are determined by the stability of the aircraft, which will be explained later

in the report. Furthermore, the angle of attack of the tail is calculated with:

$$\alpha_t = \alpha_W - i_t - \left(\epsilon_0 + \frac{\partial \epsilon}{\partial \alpha_W} \alpha_W \right) \quad (8)$$

where:

$$\alpha_W = \alpha_a = \alpha_g + \alpha_{L=0} \quad (9)$$

In this equation, α_g is the geometric angle of attack of the main wings, which is equal to 2° during cruise. Furthermore, $\alpha_{L=0}$ is the angle of attack where the main wing does not produce lift. This value can be calculated using XFOIL and is found to be -4.25° as shown in Fig.3. This gives that, during cruise, the absolute angle of attack of the main wings is equal to 6.25° . Furthermore, in Eq. 8, the tail setting angle i_t is set to 3° , the downwash at an absolute angle of attack of zero ϵ_0 can be assumed to be zero and the downwash slope $\frac{\partial \epsilon}{\partial \alpha_W}$ can assumed to be 0.42 as shown in [4]. The result is an angle of attack equal to 0.63° for the tail during cruise.

With this angle, the drag produced is 3.13 N, determined by Xfoil. Similar to the drag of the main wings, the drag of the horizontal stabilizer changes when the package is dropped. The variable that changes is in this case the geometric angle of attack α_g . As explained, this value changes from 2° to 0.95° . As a result, the angle of the tail α_t changes from 0.63° to 0.03° . This gives that the drag from the horizontal stabilizer is reduced to 2.88 N when the package is dropped.

Dimensions of the vertical stabilizers, are set to 0.35 m in chord length and the span is set to 0.4 m to accompany the fuselage parameters. For an angle of attack of 0° , XFOIL, is used to obtain a drag of 1.55 N. This value does not change when the package is dropped. As a result, the total drag of the empennage is equal to 4.68 N during cruise and 4.43 N when the package is dropped.

4.2 Total drag

With the drag values of each individual component known, the total drag can be calculated. This total drag is equal to 41.1 N during cruise and 37.3 N when the package is dropped. From this total drag and the total lift generate by the wings, a lift over drag plot can be made. This plot can be seen in Fig. 4 and shows that the maximum lift over drag ratio during cruise equals 14.6 at an angle of 3.7° . In theory, flying at this angle of attack would be the most efficient. However, when flying at this angle, the landing distance increases

by a significant margin. Therefore, the decision has been made to continue flying with an angle of attack of 2° . This results in a lift over drag ratio equal to 13.8.

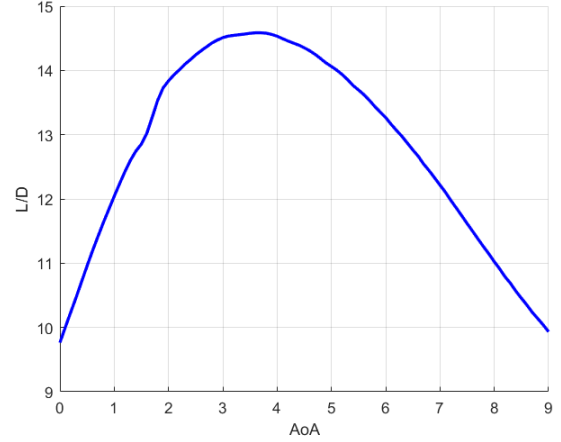


Figure 4: Lift over drag ratio for geometric angles of attack.

4.3 Power required

4.3.1 Power for takeoff

In order to choose a propulsion system, the maximum power required should be known. As maximum power is needed during takeoff, the propulsion system should be based on this value. One of the requirements is that the takeoff distance should be no longer than 150 m. Therefore, the chosen propulsion system should be able to deliver enough power such that the aircraft can take off within this takeoff distance. The takeoff distance can be calculated using:

$$S_{LO} = \frac{(1.2W)^2}{\rho_{\infty} g S C_{L,max} [T - D - \mu_r(W - L)]_{avg}} \quad (10)$$

where all values are taken at sea level. Furthermore, the thrust T , drag D and lift L are taken at the average takeoff velocity. The lift of the main wing is calculated at an angle of attack of 2° because it was chosen to have the wings inclined while the aircraft itself is horizontal. The average velocity equals $0.84 \cdot V_{stall}$ where V_{stall} is calculated using:

$$V_{stall} = \sqrt{\frac{W/S}{\frac{1}{2}\rho_{\infty} C_{L,max}}} \quad (11)$$

The maximum lift coefficient $C_{L,max}$ is calculated using XFOIL and is equal to 1.53 at an angle of

attack of 14° . Using this lift coefficient, a stall velocity of 27.6 m/s is obtained. As a result, the average velocity used for the landing distance equals 23.2 m/s. This gives that for a takeoff distance of 142.3 m, the average thrust available during takeoff should be equal to 606 N. This equals to a power of 14080 W. Therefore, the power output of the propeller should be at least 14080 W.

4.3.2 Power for Climb and Cruise

Other parameters that are necessary to calculate before a propulsion system is chosen, are the power required for climb and the power required for cruise. During level flight, it can be assumed that the thrust required should equal the total drag of the aircraft. Therefore, to calculate the power required for level flight, the total drag of the aircraft at a given altitude will be multiplied with the velocity.

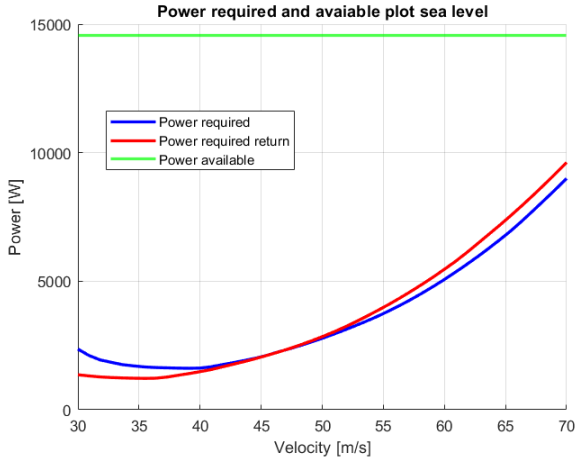


Figure 5: Power curves for flight at sea level.

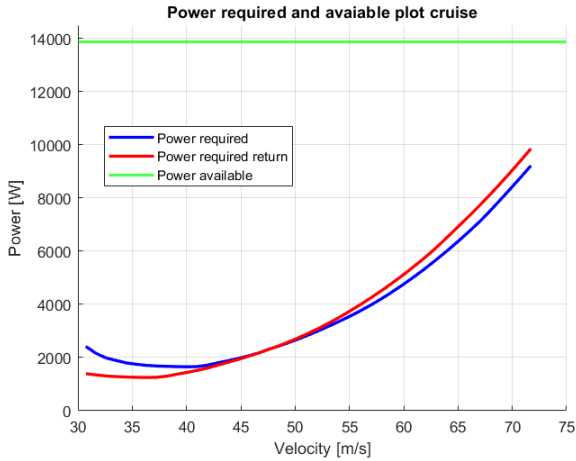


Figure 6: Power curves for flight at cruise altitude

Calculating the power required at sea level and

cruise altitude for velocities between 30 m/s and 70 m/s results in Fig. 5 and Fig. 6 where the blue lines and the red lines indicate flying with and without payload, respectively. The calculations include the change in angle of attack and therefore drag when flying at different speeds. Furthermore, the lowest velocity in the plot is 30 m/s, which is just above the stall velocity of the aircraft.

To calculate the power required for climb, first, a desired climb angle needs to be chosen. This angle is chosen to be equal to 8° . The reasoning behind this, is that with an angle of attack for the main wings of 2° during level flight, the total angle of attack during climb becomes 10° . Therefore, it is assured that the stalling angle is not reached. Furthermore, the climb velocity is set equal to 140 km/h $\approx$ 39 m/s. Around this position, the power required for level flight has its minimum and therefore the excess power necessary for climb will have its maximum. From the chosen climb angle, the rate of climb can be determined with:

$$RC = V_\infty \cdot \sin(\theta) \quad (12)$$

A result in a rate of climb equal to 3.59 m/s is obtained. For this calculation, a headwind of 50 km/h has been taken into account. Knowing the desired rate of climb, the required power for climb can be calculated using:

$$Excess\ power = RC \cdot W \quad (13)$$

This results in an excess power of 5030 W required to climb at a rate of 3.59 m/s. From Figure 5 and Figure 6, it can be seen that a mean power required of 1725 W is necessary to remain in steady flight. Therefore, the power required for climbing is equal to 6755 W. Furthermore, when the package is dropped, the climb velocity is reduced to 36 m/s where the mean power required for level flight equals 1235 W. When taking the headwind into account, the excess power required to climb after the package is dropped is calculated to be 3562 W resulting in a climb rate of 3.07 m/s. As a result, the power required for climbing without payload equals, 4797 W. The determined power requirements, indicate the properties of the electric motor and a propeller.

4.3.3 Thrust required for cruise

Fig. 7 shows the thrust required for level flight at cruise altitude for a velocity range between 30 m/s and 70 m/s. From this figure, it follows that the optimal cruising velocity for the aircraft is at 41.5

$m/s \approx 150 \text{ km/h}$ and at $33.6 \text{ m/s} \approx 121 \text{ km/h}$ for flying with and without payload, respectively. This velocity can be obtained by taking the velocity at which the minimum amount of thrust is required, as this will result in the longest range. Because the previously chosen cruise velocities equal the now defined optimal cruise velocities, these cruise velocities will still be used for further calculations.

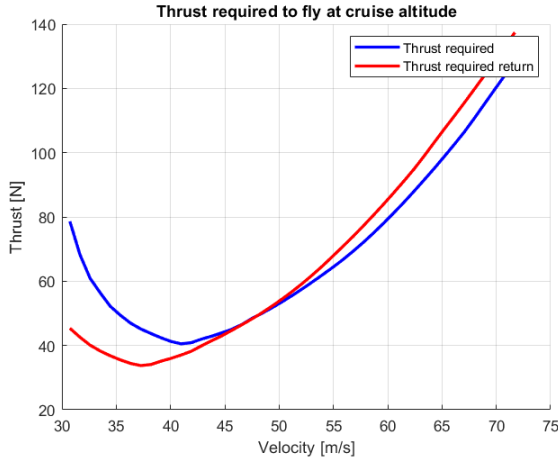


Figure 7: Thrust required curve for flight at cruise altitude

4.4 Propulsion

4.4.1 Electric motor

For the electric engine, the following requirements are set:

- High peak efficiency $> 90\%$
- RPM about 5000 for sufficient power
- Max 20 kg
- Max power output $> 14080 \text{ W}$ (takeoff power)

With these requirements, the BLDC Motor - HPM10KA/KL was found. A neodymium magnet three-phase motor with a peak power of 20 kW and maximum efficiency at 4800 RPM of 91.3% and a weight of 17 kg [9].

4.4.2 Propeller

With the known motor and power curves, a propeller can be chosen. The propeller has a maximum diameter of 850 mm due to the height of the fuselage and the maximum efficiency point of the propeller should be at cruise speed of 42 m/s. The propeller will be tuned to achieve this goal. The other parameter that will be taken into account

is the power at takeoff velocity, which is 33 m/s, should be higher than 14080 W.

With the propeller calculator provided, two propellers matched the specifications. These are the Günther 3 inch by 4.3 inch with Clark - y 5.9% and the GWS Slowfly 12 inch times 8 inch. For the Günther to give enough power, 6 blades are required for 2% more efficiency compared to the SlowFly. Due to additional weight and costs for the Günther, the GWS Slowfly with a Clark - y airfoil was chosen with 80% efficiency and 20.5 kW power at 4800 RPM. The configuration and performance curve are given in Fig. 8.

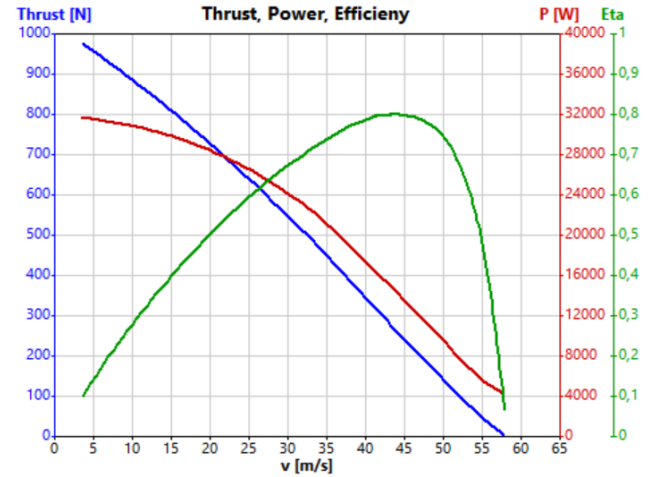


Figure 8: Propeller performance curve.

With the 91% efficiency of the engine, the total efficiency of the propulsion system equals 73%. The weight of the propeller blades is estimated by the average chord. This value is obtained by taking the mean of 10 chord values from 0 chord to 1 chord of the propeller. This results in an average chord of 0.074 m, a thickness of 0.0082 m and is calculated by multiplying the chord with 11%. The area of the propeller is estimated with the formula: $\frac{1}{2} \cdot c \cdot t$, resulting in a volume per propeller of 0.12 dm^3 . With four propellers and the material chosen to be Al2024, with a density of 2780 kg/m^3 , the weight of the propellers is calculated to be 1.4 kg.

4.5 Battery

To calculate the weight of the battery, first a battery type has to be chosen. After research, a lithium polymer battery (LiPo) was chosen due to its high discharge potential under load, high energy density and the ability to maintain low temperature at high discharge rate, making the battery safer [10]. With an energy density of 150

Wh/Kg and 300 Wh/L, the worst case that could be found was taken [11].

The total energy necessary for the entire flight is determined by summing the climb, cruise, descend and loitering energies. These energies will be determined in the following sections. Also, in these calculations, the worst case is taken into account. Therefore, as mentioned in the requirements, a headwind of 50 km/h is present. With this, it is assumed that the free stream velocity V_∞ during headwind stays equal to the free stream velocity when there is no wind. Though, the velocity relative to the ground does decrease by the wind velocity of 50 km/h. This way, the power required does not change when there is a headwind, but the flight time changes. Furthermore, to take the worst case scenario, it is assumed that the aircraft has to return from the dropping location without having dropped the package. This gives that the aircraft is heavier and therefore uses more power.

4.5.1 Energy for Climb

In one round trip, the airplane climbs to cruise altitude, descends before the drop location, climbs back to cruise altitude after having dropped its payload and descend to the runway as shown in the flight profile in Fig. 1. In this calculation, it is assumed that the drop altitude equals the takeoff and landing altitude meaning sea level, so that the worst case situation is accounted for. A climb velocity of 39 m/s has been set because, as explained, it offered the best performance and depending on the cruise altitude the time to climb will be determined.

With an average rate of climb of 3.59 m/s that was determined in Section 4.3.2, and a cruise altitude of 500 m, the time to climb equals 139 s. Using the climbing velocity of 39 m/s with a 50 km/h headwind, the distance to reach cruise altitude is calculated to be equal to 3.47 km. Furthermore, using the energy necessary to climb with a headwind of 50 km/h, defined in Section 4.3.2, and the climbing time of 139 s, the energy required for climbing can be calculated. This results in 0.94 MJ required for one climb, and therefore 1.88 MJ for the two climbs per flight. Here it is once again assumed that the second climb, after dropping the payload, also starts at sea level.

4.5.2 Energy for descend

For descending, it is assumed that the rate of descend will be equal to the rate of climb. As a result, the distance used for descending will also be equal to the distance used for climbing when assuming equal velocities. Furthermore, during the descending it is assumed that no power is used, and therefore, descending does not contribute to the battery weight.

4.5.3 Energy for Cruise

The energy necessary for cruise is once again determined by taking a headwind of 50 km/h into account. The total flying distance at cruise altitude is equal to the distance from the takeoff location to the dropping location, minus the climbing and descending distances. The total flying distance from the takeoff location to the dropping location, in the worst case scenario, is equal to 50 km plus 10 km additional distance. As a result, the distance travelled at cruising altitude will be equal to $60 - 2 \cdot 3.47 = 53.06$ km. Because it is assumed that the altitude at which the package is dropped equals the takeoff altitude, the entire flight will consist of two equal sections, hence the distance is doubled to a total cruising distance of 106.12 km.

Taking into account the headwind of 50 km/h, the cruise velocity relative to the ground equals $100 \text{ km/h} \approx 27.8 \text{ m/s}$. Using this cruising distance and relative cruising velocity, a cruising time of 3820 s is found. With the power required during cruise at 500 m being equal to 1830 W, the total energy consumed during cruise equals 6.99 MJ.

4.5.4 Energy for Loitering

The loitering velocity is set to 36 m/s (minimal power required) to achieve maximal endurance. The loitering time is equal to the requirement of 15 minutes or 900 s. The loitering altitude is at cruise level, since this is already very low. The power required to fly at 500 m with a velocity of 36 m/s equals 1210 W and therefore the energy required for loitering is 1.1 MJ.

4.5.5 Total energy for flight

Adding the energies listed above results in a total energy needed, assuming worst case scenario, of 9.97 MJ. With the motor and propeller efficiency and a safety factor of 1.4 the total battery energy required is 19.2 MJ or 5.32 kWh. A battery weight

of 35.5 kg and volume of 17.7 L are adequate sizes for this power requirement.

To summarize the used energy per stage of flight, Fig. 9 shows the required energy to fly at a certain altitude. In this graph, the energy required for the cruise and climb are plotted for one climb and 60 km of flight distance. For the total energy required, these two values should be doubled.

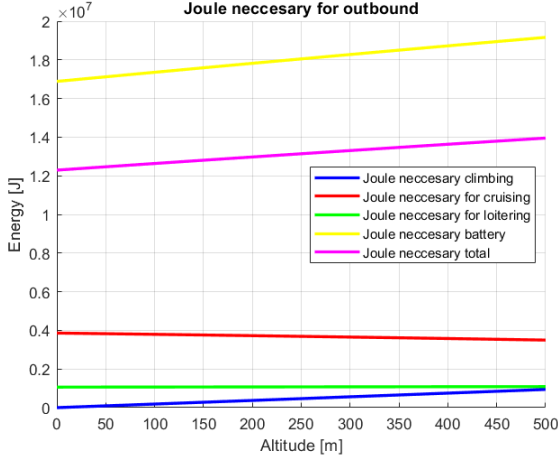


Figure 9: Required energy

From this graph, it can be seen that an altitude of 500m that has been set as the cruising altitude, does not pose significant changes on the required energy.

4.6 Landing performance

In the requirements, it is stated that the landing distance of the aircraft should be less than 150 m. To check this, the landing distance needs to be calculated with [4]:

$$S_L = \frac{(1.3W)^2}{\rho_{\infty} g S C_{L,max} [D + \mu_r (W - L)] 0.91 V_{stall}} \quad (14)$$

Again the worst case scenario is taken into account, therefore, the weight still includes the weight of the package. Furthermore, the same values are used as for the takeoff performance, although the forces are calculated at $0.91 \cdot V_{stall}$. Furthermore, the friction coefficient μ_r is assumed to be 0.04 without brakes and 0.44 with brakes [4]. The result is a landing distance of 1189 m without brakes and 143 m with brakes. For this reason, disk brakes on the wheels are recommended, as stated in the overview.

4.7 Stability

In this stability analysis, the longitudinal stability of the aircraft will be analyzed. For an aircraft to be called longitudinal stable, the aircraft should have a positive moment about its center of gravity when the angle of attack is higher than the equilibrium angle of attack (α_e). Because of this positive moment, the aircraft will pitch back down to its equilibrium angle of attack α_e . Vice versa, the the aircraft should have a negative moment about its center of gravity when the angle of attack is lower than the equilibrium angle of attack (α_e). This will then pitch the plane back up, resulting in a stable flight. Concluding, to ensure stable flight, the moment coefficient slope should have a negative value and the zero-lift moment coefficient should have a positive value.

4.7.1 Center of gravity

To perform a stability analysis, it is important to know the position where the resulting moments are acting on. Because this position is known as the center of gravity, first, the position of this center of gravity should be calculated. To calculate this position, the weight of the individual components will be assumed to be equal to a point mass. Thereafter, a sketch can be created showing all the relevant point masses and their position. This sketch can be found in Fig. 32 in Appendix 12. Furthermore, the location of the center of gravity can be calculated by [4]:

$$x_{cg} = \frac{\sum (x_i M_i)}{\sum M_i} \quad (15)$$

where, x_i is the position of component i and M_i is the mass of component i. However, when the package is dropped from the airplane, the position of the center of gravity can change. In this design, it is chosen to place the package in the aircraft such that the position of the center of gravity does not shift when the package is dropped. This makes the design more simple and ensures stability in both cases, when the package is inside the fuselage and when the package is dropped. To achieve this, first, the center of gravity is calculated without including the package. Using Eq. 15 and the values shown in Figure 32 in Appendix 12, gives that the center of gravity without package is located at $x = 0.58$ m. If the package is placed at the same location as the center of gravity, the center of gravity will remain at the same location, whether the package is dropped or not.

Therefore, the center of gravity with the package is also located at $x = 0.58$ m.

4.7.2 Analysis

As explained, to determine whether the aircraft is longitudinal stable, the moment coefficient slope and moment coefficient at zero lift need to be known. The moment coefficient slope can be calculated using:

$$\frac{\partial C_{M,CG}}{\partial \alpha_a} = a_W \left[h - h_{ac,W} - V_H \frac{a_T}{a_W} \left(1 - \frac{\partial \epsilon}{\partial \alpha_W} \right) \right] \quad (16)$$

and the moment coefficient at zero lift can be calculated using:

$$C_{M,0} = C_{M,acwb} + V_H a_T (i_t + \epsilon_0) \quad (17)$$

In the equations, the lift slope of the main wing a_W and tail a_T are calculated using XFOIL, which both equal to 0.11. With a Reynolds number of $1.37 \cdot 10^6$ and a Mach number of 0.1234, the moment coefficient about the aerodynamic center of the wing $C_{M,acwb}$ is calculated to be equal to -0.105. Furthermore, the tail volume ratio can be calculated using:

$$V_H = \frac{l_t S_t}{S \cdot c} \quad (18)$$

where l_t is the distance between the center of gravity of the aircraft and the aerodynamic center of the tail, S_t is the surface area of the tail, S is the surface area of the main wing and c is the chord length of the main wing. l_t can be calculated by subtracting the position of the aircraft's center of gravity from the aerodynamic center of the tail. This results in an l_t of 1.3 m. Furthermore, the chord is equal to 0.5 m, and the surface area is equal to 1.96 m^2 , both showed in Table 2. The chord of the horizontal stabilizer was assumed to be 0.35 m and the length of the horizontal stabilizer was assumed to be 1.5 m, which gives a surface area of the tail S_t of 0.525 m^2 . This gives a tail volume ratio V_H of 0.69. The tail setting angle i_t and downwash at zero lift ϵ_0 in Eq. 17 are assumed in Section 4.1.6 to be equal to 3 degrees and zero, respectively. Moreover, the downwash slope $\frac{\partial \epsilon}{\partial \alpha_W}$ is in this section also assumed to be equal to 0.42 [4] and the distance between the center of gravity of the aircraft and the aerodynamic center of the main wing, scaled with the chord, $h - h_{ac,W}$, is calculated to be 0.25. This, in the end, gives that the moment coefficient slope is equal to -0.0073 and the moment coefficient at

zero lift is equal to 0.1183. As stated before, this will result in a longitudinal, stable aircraft. To plot these values, the following equation is used:

$$C_{M,CG} = C_{M,0} + \frac{\partial C_{M,CG}}{\partial \alpha_a} \alpha_a \quad (19)$$

In this equation, the moment coefficient at zero lift and the moment coefficient slope have been calculated in this section and the absolute angle of attack α_a has been calculated in Eq. 9 to be equal to 6.25° . Plotting this equation results in the blue line visualized in Fig. 10. As can be seen from this figure, the blue line is not crossing the zero moment coefficient line. This means that there is no angle of attack within this plotted range for which the aircraft is trimmed. To trim the aircraft to the right angle of attack, the elevator on the horizontal stabilizer is deflected. This deflection angle can be calculated using:

$$\delta_{trim} = \frac{C_{M,0} + (\partial C_{M,CG} / \partial \alpha_a) \alpha_a}{V_H (\partial C_{L,t} / \partial \delta_e)} \quad (20)$$

where $\partial C_{L,t} / \partial \delta_e$ is assumed to be 0.04 [4] and other values were already given or calculated. This gives as a result that the elevator deflection δ_{trim} should equal 2.63° . Because of the positive value, the elevator will be deflected downwards. This results in extra lift that is being generated by the horizontal stabilizer. As a result of the elevator angle, the blue line will translate into the red line. This line shows, that the aircraft is trimmed for an absolute angle of attack of 6.25° .

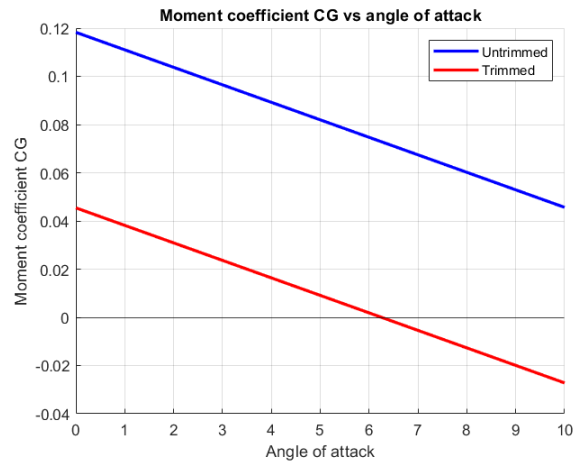


Figure 10: Moment coefficient about CG vs angle of attack

An alternative stability criterion is the concept of the neutral point. To assure longitudinal stability, the position of the center of gravity of the aircraft

should always be forward of the neutral point. The neutral point can be calculated using:

$$h_n = h_{acwb} - V_H \frac{a_T}{a_W} \left(1 - \frac{\partial \epsilon}{\partial \alpha_W} \right) \quad (21)$$

which results in the neutral point, scaled with the chord, calculated to be 0.63. The static margin can then be calculated using $h_n - h$. This results in a static margin equal to 0.06. Because this value is positive, the neutral point lies behind the center of gravity of the aircraft, resulting in longitudinal stability.

5 Structural Analysis

Through this section, the structure of the aircraft is described. Starting with the fuselage and then the structure of the wing, the optimal number of supports is chosen for structural stability.

5.1 Fuselage Structure

The fuselage consists of stringers, frames, longerons and a skin. The structural components that require high strength such as the stringers, frames and longerons are made from Aluminum 7075 [12]. Aluminum 2024 was chosen for the fuselage, as it has a high resistance against high cyclic fatigue [12]. The properties of both materials are shown in Tables 6 and 7 in Appendix 12.

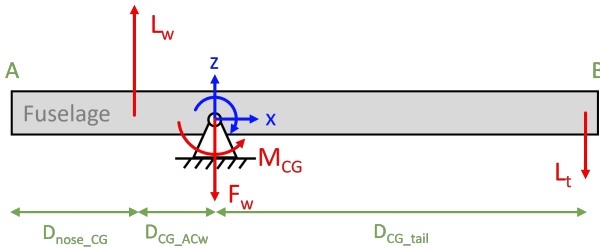


Figure 11: Free body diagram of the fuselage.

Fig. 11 shows a schematic free body diagram (FBD) of the aircraft. Simplifications are made, such as the assumptions that the fuselage is a large, slender-member beam and that the weight is concentrated at the centre of gravity. Caused by the bending, the top part of the fuselage experiences tensile stresses, while the bottom part of the fuselage is under compression.

The combination of the gravity force, lift of the wing and the lift of the tail cause internal shear forces and moments, these are shown in Figure 12. It can be seen that the maximum bending moment

is -2.30 kNm, located at the centre of gravity of the plane.

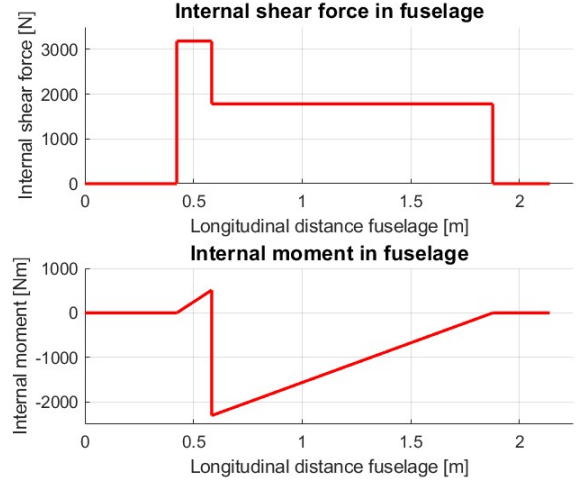


Figure 12: Internal shear forces and moments along the fuselage.

5.1.1 Bending stresses

To determine the maximum stresses in the fuselage due to the bending moment, the following equation was used [13].

$$\sigma_{max} = \frac{M(x)y}{I} \quad (22)$$

Where $M(x)$ is the maximum bending stress in the fuselage, y is half the height of the fuselage and I is the second moment of area. The fuselage was assumed to be a hollow rectangle of $0.8 \times 0.6 \times 0.0015$ m ($w \times h \times t$), with $I_{fuselage} = (wh^3)/12 - (w - 2t)(h - 2t)^3/12 = 1.43 \cdot 10^{-4}$ m⁴. The thickness of 0.8 mm was chosen to ensure local buckling does not occur. Four longerons with the shape of an I-beam where added to the corners, as attachment points for the package and battery. Table 8 in Appendix 12 shows their dimensions, this results in a combined moment of inertia of $I_{fuselage,total} = 2.11 \cdot 10^{-4}$ m⁴. With a safety factor of 1.5, the resulting maximum stress is 3.26 MPa. In the next sections, this value will be compared with the critical global and local buckling stresses to determine if buckling will happen and what the stringer and frame spacings should be.

5.1.2 Stringers

The shape of the stringers was chosen to be a z-section, for its high stiffness and easy access for attaching it to the skin [14]. In addition, a z-shape is easy and cheap to manufacture. Eq. 23

[13] is used to calculate the critical stress for local buckling.

$$\sigma_{crit} = \frac{k\pi^2 E}{12(1-\nu^2)} \left(\frac{t}{b}\right)^2 \quad (23)$$

The buckling coefficient k for clamped loaded and unloaded edges is 7.5 as given in [13]. With the materials properties and stringer dimensions mentioned beforehand, a stringer spacing of 0.08 m, combined with a thickness of 0.8 mm result in a critical stress $\sigma_{crit} = 50.6$ MPa. That is higher than the maximum applied stress of 3.26 MPa, therefore a total of 35 stringers are needed for the entire fuselage.

5.1.3 Frames

Frames increase the structural integrity of the fuselage and its ability to withstand global buckling, where both the skin and stringers buckle under compressive loads. The highest compressive stresses occur at the bottom of the fuselage. The spacing of the frames can be determined by the critical global buckling stress, which is given by the following Eq. 24, [13].

$$\sigma_{crit,global} = \frac{\pi^2 EI}{Al^2} \quad (24)$$

Where E is the Young's Modulus, I is the inertia, l the spacing of the ribs and A is the area of the stringers and skin. Here a plate section of the fuselage between 5 stringers and 2 ribs is used. With the dimensions of the stringers found in Table 9 in Appendix 12 the inertia $I = 9.90 \cdot 10^{-9} \text{ m}^4$. The area of both the skin and 5 stringers was calculated to be $A = 5.48 \cdot 10^{-4} \text{ m}^2$. Inserting these parameters in Eq. 24 results in a critical stress of 315.9 MPa with a frame spacing of 0.45 m, which will resist global buckling. In total 3 ribs are needed for the fuselage.

5.1.4 Tail rods

Two tail rods bridge the distance between the empennage and the fuselage. The maximum bending moment in the rods is -1.31 kNm as shown in Fig. 12, which occurs at the end of the fuselage. A hollow tube is chosen, with an outer diameter of 8 cm and an inner diameter of 7 cm. Using Eq. 22 a maximum bending stress of 31.6 MPa was calculated. With $I_{rod} = \pi((D_{outer}/2)^4 - (D_{inner}/2)^4)/4$. Fig. 13 shows the deflection curve of the one of the tail rods, caused by the internal

moment. It shows a maximum deflection of 26.9 mm at the location of the empennage.

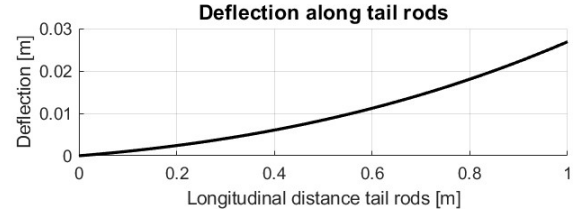


Figure 13: Deflection tail rods.

5.2 Wing Structure

The wings are build up from a spar, ribs, stringers and the skin. The components that are under high stress due to the lift force, require a material with a high yield strength. The spar, which is an I-beam, the ribs, stringers and the upper skin of the wing are the same AL7075-T6 as is used in the fuselage [15]. For the skin on the lower surface of the wings, AL2024 [16] is used, because it can handle the tensile stresses as well, and is cheaper. In the whole section below about the wings, only one side is taken into account to do all the calculations on, because both sides are equal and thus have the same values, except sometimes a different direction of forces. Fig. 14 shows the FBD of the used case.

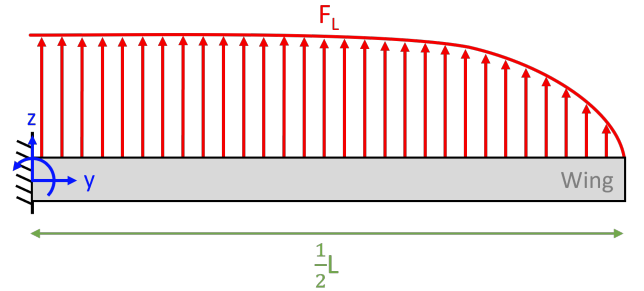


Figure 14: Free body diagram of one wing.

5.2.1 Internal Forces

From the FBD, the internal shear force and moment are calculated with Newton's second law, from the distributed lift force function. This lift distribution function is from the combination of the *Kutta-Joukowski* condition with the *elliptical circulation distribution* [17] [7]. The elliptical circulation distribution can be used for inviscid incompressible flow at low speeds, as is the case here. Consider the following:

$$l(y) = \frac{2W}{b} \sqrt{1 - \left(\frac{y}{b}\right)^2} \quad (25)$$

Where W is the weight in Newtons and b is $1/2L$. The reaction force R_y is simply the integral of Eq. 25 from 0 to the span. This results in a total reaction force of 2.20 kN. The internal shear force function is now the reaction force minus the primitive of the force distribution function and then calculated at location y .

$$S(y) = R_y - \frac{W(b \cdot \arcsin(\frac{y}{b}) + y\sqrt{1 - (\frac{y}{b})^2})}{b}$$

Where b is the length of one wing and W the weight of the plane in Newtons. The resulting moment is the integral of the shear force function above, leading to a moment of 1.83 kNm. The moment function is the integral of the shear force function.

$$M(x) = M_0 - R_y \cdot y + \int_0^y l(y) \cdot \left[\int_0^{y_c} l(y) = \frac{\int_0^y l(y)}{2} \right]$$

The function between brackets is solved numerically to locate the connecting location of the distributed force. The graphs of Fig. 15 are the result.

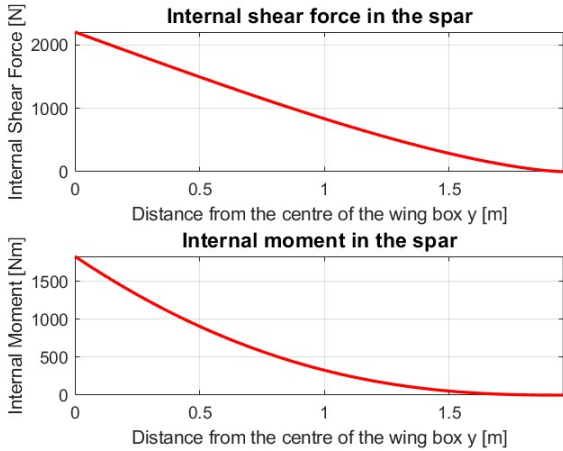


Figure 15: Internal shear and moment diagram.

5.2.2 Spar

The spar is the main structural component of the wing. It is subjected to all of the stresses caused by the lift force. To minimize its weight, but still have a large second moment of area, the spar will be an I-beam. Rewriting the second moment of area of an I-beam, its minimal width can be calculated. The maximum stress in the spar occurs at its connection point with the fuselage. The maximum stress the material can hold before plastic deformation is its yield stress. The maximum stress applied is the yield stress divided by a safety

factor, taking into account the lifespan of the airplane. From the S-N curve, and given that during one flight, two load cycles are applied (take-off and landing) and the lifespan is at least 1000 flights. From Fig. 16, the applied stress at 267.6 MPa leads to a minimum of 2000 safe flights before the change of fatigue failure. This gives a safety factor of 1.7.

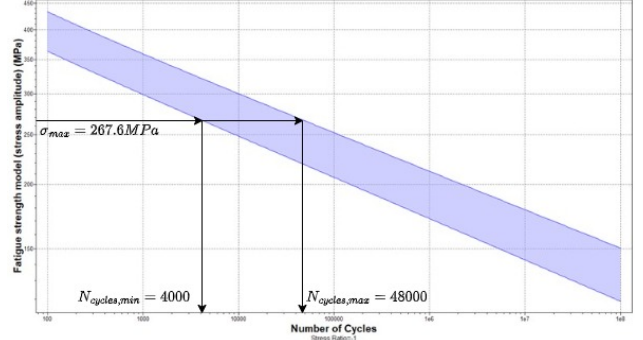


Figure 16: S-N range for AL7075-T6.

Rewriting Eq. 22, and then filling in the second moment of inertia, the minimal width can be found.

$$I_{min} = \frac{M(0) \cdot H}{2\sigma_{applied}} \quad (26)$$

Where $M(0) = 1.83$ kNm, H is the height of the I-beam which needs to fit inside the wing. The maximum height of the wing is $0.12 \cdot chord = 60$ mm. Because the I-beam will be straight and the wing is not, The maximum height is assumed at 50 mm. From the second moment of area equation of an I-beam.

$$w_{min} = \frac{12I_{min} - t \cdot (H - 2t)^3}{H^3 - (H - 2t)^3} \quad (27)$$

The thickness is assumed to be 3 mm and then the minimal width is 45.2 mm.

To find the deflection of the beam, the force distribution is now assumed to be a linearly variable load and still is zero at the tip. This assumption is made to use some known formulas for the deflection and angle.

$$v(y) = \frac{l(0) \cdot y^2}{120bEI} (10b^3 - 10b^2y + 5by^2 - y^3) \quad (28)$$

$$\theta(y) = v'(y) = \frac{l(0) \cdot y^2}{24bEI} (4b^3 - 6b^2y + 4by^2 - y^3)$$

To find the axial force on the skin, also the curve function is computed.

$$\kappa(y) = v''(y) = \frac{l(0) \cdot y^2}{24bEI} (8b^3 - 18b^2y + 16by^2 - 5y^3)$$

All these functions are shown in Fig. 17. This gives a maximum displacement of 57.4 mm and an angle of 0.0366 rad.

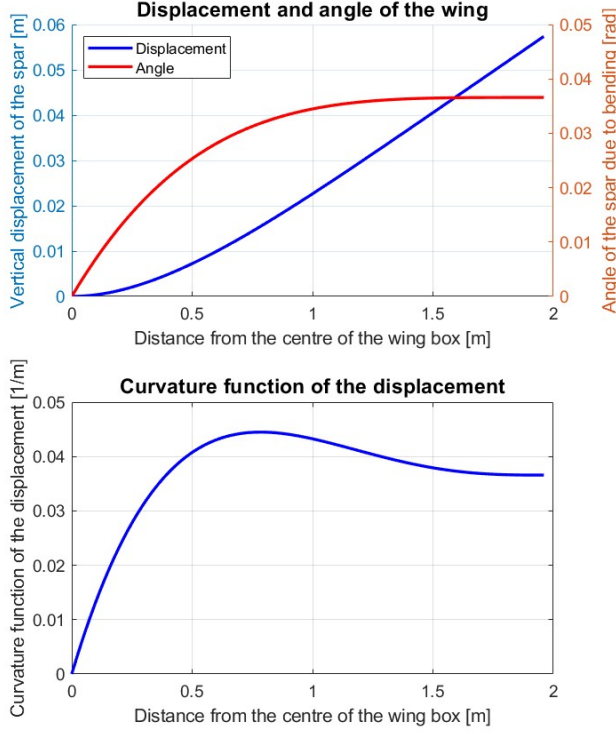


Figure 17: Vertical displacement, angle and curvature of the spar.

5.2.3 Stringers

The deflection of the spar, and thus the wing, causes an axial compression force on the top of the wing. That could cause buckling of the skin, which is something that needs to be avoided. Therefore, stringers are added onto the upper skin to prevent buckling. The stringer type is a very standard L shape stringer with both sides 20 mm long and 1.5 mm thick. To calculate the spacing, the axial stress applied needs to be lower than the critical buckling stress. Assumed is that all edges of the plate are simply supported, thus, $k_{min} = 4$ and the safety factor (SF) is 1.6. The height from the neutral line is assumed to halve the height of the wing, thus $z_{max} = 0.5 \cdot 0.12c = 30$ mm.

Filling in the following equation with the parameters above, $t = 1.5$ mm, ν is Poisson ratio, which is 0.33 for this material, $\kappa_{max} = 0.0445$ 1/m and the chord is 0.5 m. This leads to a stringer spacing of 62.4 mm that is located on the upper surface of the wing.

$$\sigma_{applied} \cdot SF = \sigma_{cr}$$

$$SF \cdot E_{7075-T6} \cdot \kappa_{max} \cdot z_{max} = \frac{k \cdot \pi^2 \cdot E_{7075-T6} \cdot t^2}{12L^2 \cdot (1 - \nu^2)}$$

$$L_{stringerspace} = \sqrt{\frac{k \cdot \pi^2 \cdot t^2}{SF \cdot 12(1 - \nu^2) \cdot \kappa_{max} \cdot z_{max}}}$$

For the lower side, which is in tension, stringers are only needed if the tensile stress applied is larger than the yield stress divided by a safety factor. The tensile stress applied is 97.6 MPa and the yield stress of Aluminium 2024 [16] is 324 MPa, so well within the limit. This means that stringers are not needed here.

5.2.4 Ribs

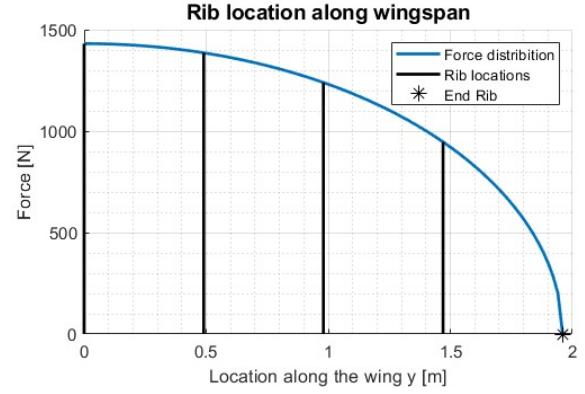


Figure 18: Rib location wing.

The force of the stringers needs to be transferred to the spar. Ribs are placed along the spar in the shape of the airfoil. These spars are also I-beams but with a small width just to mount the skin on. To calculate the spacing between the ribs, the case of global buckling is applied.

$$F_{applied} \cdot SF = P_{cr}$$

$$A_{wing} \cdot E_{7075-T6} \cdot \kappa_{max} \cdot z_{max} = \frac{\pi^2 E_{7075-T6} \cdot I_{wing,min}}{L_{ribspace}^2}$$

$$L_{ribspace} = \sqrt{\frac{\pi^2 \cdot I_{wing,min}}{A_{wing} \cdot \kappa_{max} \cdot z_{max}}}$$

Filling in this equation with a wing area of $c \cdot z_{max} = 0.015$ m² and $I_{wing,min} = 9.426$ m⁴, which is the second moment of area of the skin and stringers around the neutral axis of the wing. This gives a maximal spacing of the ribs which is 0.539

m. This means that a minimal of four spaces is needed, and thus five ribs. The distance between the ribs reduces to 0.49 m. Fig. 18 shows the spacing and the force distribution along the wing.

5.3 Turning flight

A flight envelope shows the aircraft's structural limits in terms of its various flight parameters. A parameter that is of particular interest structurally is the turn radius. A low turn radius of 100 m is desired so that the aircraft can return to base swiftly after delivering the package. The turn radius is used to determine the load factor experienced by the plane:

$$\phi = \arctan\left(\frac{v_{\text{cruise}}^2}{9.81 \cdot R}\right)$$

$$n_1 = \frac{1}{\cos \phi} \quad (29)$$

For a 100 m turn radius, the load factor is calculated to be 2.03. With this value, a flight envelope shown in Fig. 19 has been plotted, for two extreme flying altitude, sea level and the 3500 m. With this flight envelope, the safe airspeeds and flight limits can be seen.

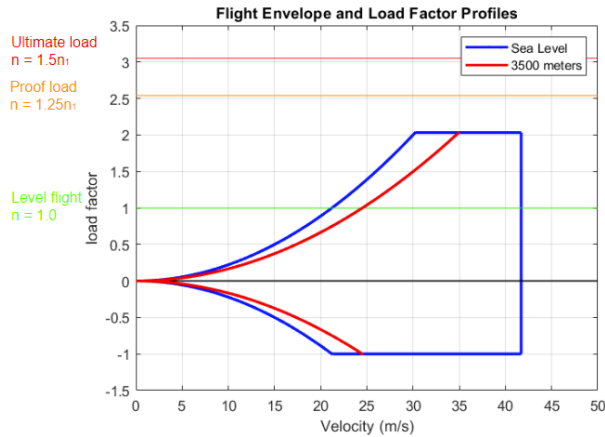


Figure 19: Flight envelope at sea level

6 Conceptual design

Fig. 20 visualizes the conceptual design with the parameters that were obtained, as discussed in previous sections. To further view the design in more detail, additional renders have been included in Section 12. The technical drawing with the most important parameters can be seen in Fig. 31.



Figure 20: Visualization of the conceptual design

7 Conclusion/Discussion

7.1 Requirements

Among other requirements, this report set out to optimize the aircraft for a cruising velocity of 150 km/h, reaching a minimum velocity of 100 km/h before stalling and a climbing rate of at least 3.0 [m/s]. As presented thus far, these requirements have been met and in the case of the climbing rate exceeded to 3.6 [m/s].

Furthermore, the structural analysis conducted indicates the aircraft ability to operate at an altitude of 3500 m and a minimum of 2000 safe flights before potential fatigue damage.

7.2 Conclusion

To summarize, this report documents the design process of an aircraft that is capable of delivering a payload of 25 kg to a maximum distance of 60 km. The performance of the design has been verified using flight mechanics and aerodynamics together with structural analysis exploring the feasibility of the design. Important findings and final dimensions of the design are included in the fact-sheet in Appendix 12 along with a CAD model of the aircraft.

While this report offers valuable insights, it acknowledges its limitations. Further investigation into specific parts such as high lift devices, landing gear and avionics. Another point of expansion is research into the energy density of Lithium batteries, which could pave the way for a growth in the market for UAV's used for disaster relief even further.

This report serves as a foundation, and invites future research into the design.

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9 Appendix A: Business case

9.1 Value Proposition

Disaster relief organization, such as *Wings for Aid*, face a problem with delivering small aid packages, to individuals or small groups in need. Whether it's due to high operational costs, availability of equipment and trained pilots, or lack of access by land or water, the current challenges in delivering small aid packages demand innovative solutions. This is where *Twente In The Air Netherlands*'s new conceptual design of a fixed wing aircraft comes into play. Tailor-made for delivering a 25kg package to disaster struck areas up to 50km away, the optimized design aims to provide an alternative unmanned solution, to supplying small groups in need through the use of affordable materials, an autonomous flight system, modular parts and easy access to the payload compartment.

9.1.1 Customer Segments

The disaster relief organization that are the primary customer for the design, have a number of methods of providing relief, specifically delivering supplies. Such a job provides valuable benefits, such as saving human life, reducing damages and preventing the disaster from spreading. Unfortunately, during a disaster response, organizations face a number of pains such as the emergency response equipment and team being preoccupied, time constraints imposed by the urgency of the disaster, transport to remote areas and high cost.

9.1.2 Value Proposition

To account for the pains experienced by the customer, the aircraft provided to the organizations would be accompanied by a basic training and instructions on how to operate the aircraft. Once in operation, the aircraft layout allows for the replacement of the battery and payload through an easy-to-use hatch located on the back of the fuselage, drastically reducing the time constraint on the crew. Furthermore, the aircraft's autonomous flight computer allows the operator to update the flight path remotely, eliminating the constant need for piloting the aircraft. This feature gives an operator the opportunity to control multiple aircraft at once. Moreover, the aircraft embodies the idea of modularity, large segments of the aircraft are detachable, like the wings and tail. Coupled with a subscription based replacement service, makes it a great for not only compact storage, but also hands-free maintenance and regular replacement. The versatility of this design makes it essentially future-proof, because it allows for add-on features, and custom parts to be attached to the convenient rail system of the aircraft, granting possibilities for use aside from payload delivery. Consider value proposition canvas, in Fig. 21.

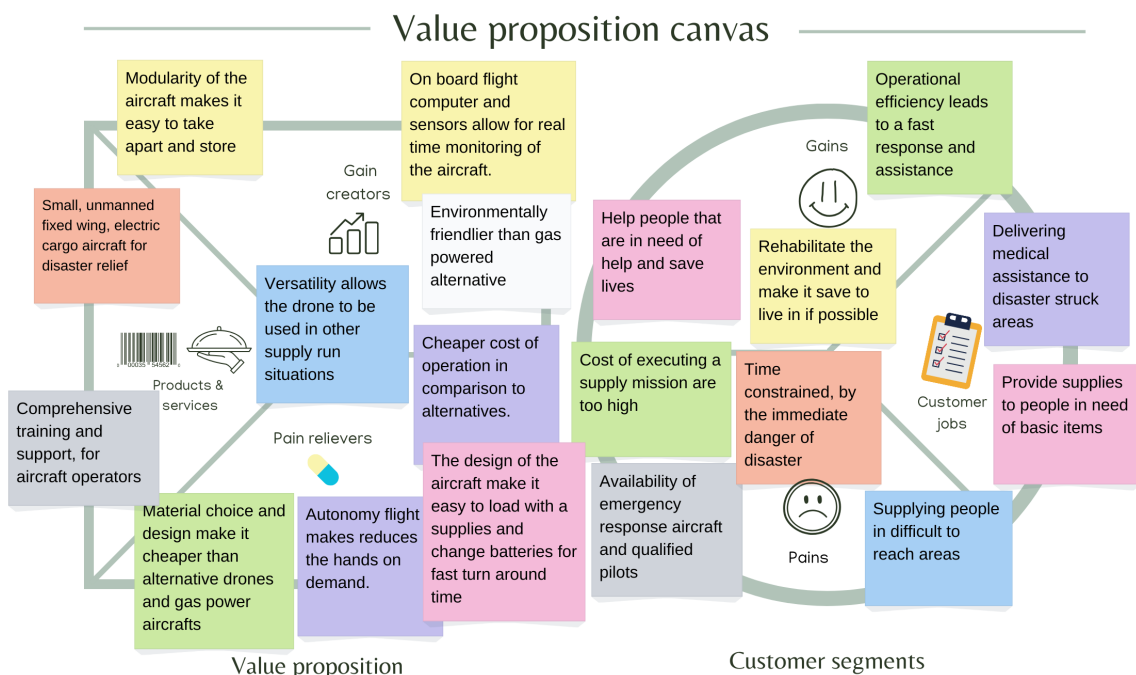


Figure 21: Value proposition canvas

9.1.3 Use Cases

Consider the following two exemplary use cases for the aircraft.

- An emergency medical supply delivery to a person located in an area where the roads are blocked, impassable or unreachable by land.
- A search and rescue support for survivors that must stay overnight in a hard-to-reach locations during in a dangerous climate, such as mountains.

9.1.4 Validation & Ideal Customer

As a validation of the value proposition, *Avinics* can be considered as a potential customer. *Avinics* is based across six European countries and provides critical aviation services to the civil sector to save lives and protect communities. With 200 helicopters in its fleet scattered across Europe, *Avinics* came to the aid of more than 65,000 patients and organized about 87,000 missions in 2021[18]. *Avinics* is an organization that can be considered a potential customer, because it specializes in disaster related, emergency medical services and search and rescue missions on a regular basis. The aircraft presented in the report is of great value to organizations such as *Avinics*, because the aircraft is much more cost-effective than helicopters. More importantly, *Avinics* could make use of two rail system's design quality by mounting a search system such as a mobile phone or avalanche beacon tracker on these rails, making it great for surveillance. Aside from administering aid, *Avinics* can use the aircraft for delivering supplies to remote villages or settlements on islands [19] that rely on helicopter flights for their supplies.

Other disaster relief organizations as shown in Table 3 would also benefit from unmanned aircraft in their arsenal, which is why the design of the aircraft has a unique value.

Organisation	Operating region	Number of active duty operators
The International Federation of Red Cross and Red Crescent Societies (IFRC)	191 countries	16 million volunteers[20]
The Dutch Relief Alliance	15 countries[21]	Not specified
Australian State Emergency Service	Australia	43000 volunteers
Americares	Global[22]	Not specified

Table 3: Potential costumers

9.2 Market analysis

9.2.1 Target market & size

The aerospace market as a whole has a size of 321.5 billion USD [23]. However, focusing on the specialized niche of the emergency and disaster

response sector within this expansive market, it is found that the market is estimated to be valued at 166.7 billion USD [24]. Within this sector, the global emergency drone market represents a distinctive subset, serving as a vital component in disaster response strategies. Even though the aircraft has a wide range of applications, it is designed with disaster relief in mind. Hence, why the 166.7 billion USD can be considered as the total available market size.

Transitioning to the serviceable available market, attention is directed to the global emergency drone market, which has a specific value of USD 4.885 billion [25]. This value underlines the portion of the total market where drones are actively utilized for disaster response. This is considered to be the primary target market of the aircraft.

To set a reasonable goal, based on the location of Airbus's headquarters, the serviceable obtainable market can be reduced to Europe which make up 31% of the global market. Consider the size of the market summarized in Fig.22.

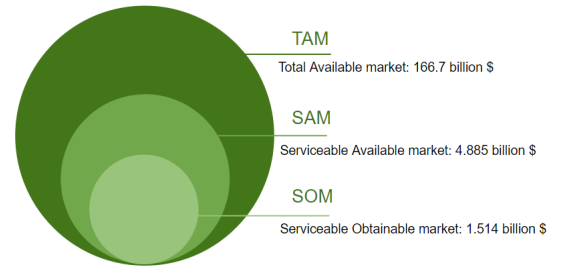


Figure 22: TAM-SAM-SOM Analysis, [24] [25]

9.2.2 Market growth

Looking at the global emergency drone market, there are two ways in which market growth can be analysed. Firstly, the increase in demand, how natural disasters are predicted to behave in the future, and secondly the technological impacts of drones and their predicted development.

According to the *Asian Disaster Reduction Center* [26], the amount of Natural Disasters in 2022 was estimated at 388, with a number of affected people reaching 190 million. This number is set to increase as the global reported number of disasters tends the increase over the years [27].

Consider the research data by *Our World* indicating the growing number in Fig. 23. As the number of disasters increase, the market for emergency and the number of disaster response units is expected to grow with it.

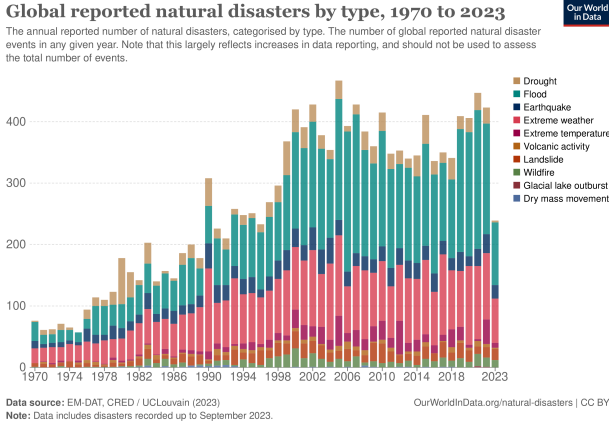


Figure 23: Types and distribution of natural disasters[27]

On the other hand, the drone market has also experienced a large surge in growth [28] in the last decade. Advancements in the aerospace sector as well as the increase of drone use in the military have lead to a high demand for aerial vehicles [29].

Combining these two market trends, a growth in the global emergency drone market is almost certain. This is confirmed by an impressive compound annual growth rate of 13.1% [25] that is expected to increase the size of the market to 16.729 billion USD by 2033.

9.2.3 Competition

An analysis of the competition can be seen in Table. 4.

Company/product	Notes
Pterodynamics X-P5[30]	Direct competition , and Military use VTOL capable drone, max payload of 23kg
Zipline [31]	Direct competition, and transport and logistics Medical supply - No product available yet
Canon Dynamics	Direct competition, Supply delivery, concept only
SYPAQ[32]	Indirect competition, transport and logistics Extremely low cost, made from cardboard
Draganfly[33]	Indirect competition broad aerospace market Wide range of applications and quad-copter

Table 4: Competition analysis

Companies that are considered a direct competition² have invested stake or plan to enter the emergency drone market in the future. The table also includes similar designs that operate in other markets and also completely different drone types like quad-copters.

9.2.4 Competitive advantage

The position of the proposed design with respect to the competition is visualized in Fig. 24. On y-axis, the scope of the market is shown, where the niche market is the emergency drone market

and the complete aerospace market is the total market. The x-axis the competitive advantage is shown, with focus on low cost and differentiation.

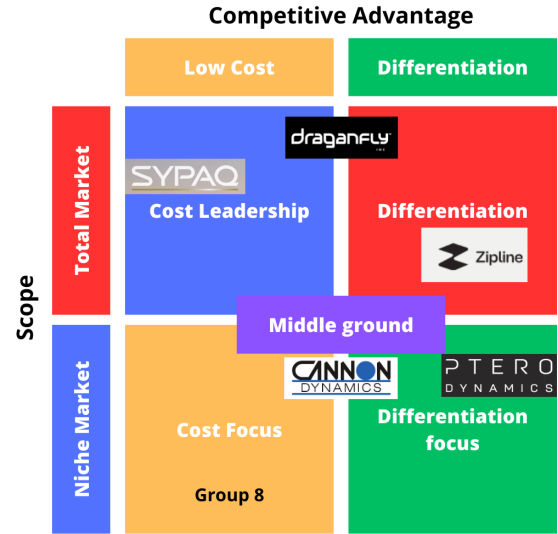


Figure 24: Porter's Generic Strategy for competition in global emergency drone market, [30] [31][32] [33]

The design of the aircraft can be seen position in the bottom left in the cost focus group, because it targets a niche market and focuses on the costs available. Consider the positions of the exemplary competition that is described in table 4. As it can be seen, the design has a unique position over the competition as it is specifically focused for disaster relief organizations. This gives it its competitive advantage that can be used to reach an adequate market share value in a rapidly expanding market.

9.3 SWOT analysis

To evaluate the design, and its place in the market, a SWOT analysis has been conducted, as shown in Fig. 33.

9.3.1 Strengths

The primary strength over other designs is the modularity provided by the rails. This rail design significantly reduces the amount of material needed, lowering the cost while providing the customer with the freedom of to customize the airplane. This is the unique value that the design provides. Consider Fig.33 for other strengths.

9.3.2 Weaknesses

The airplane's major drawback is navigation in mountains terrain. Due to the maximum flight altitude being 3500 [m], VTOL aircraft could bene-

²Company which is invested/interested in the emergency drone market

fit in certain situations. Since the airplane is unmanned, it has not been optimized for stability and is susceptible to weather conditions. Lastly, the current design also does not have any countermeasures for attacks, making it vulnerable.

9.3.3 Opportunities

Opportunities in the aircraft present themselves in the form of collaboration. As the market is growing from an increased number of disasters, news and old organizations will seek additional features that the aircraft can accompany using the air system. Through this modular rail design, the aircraft can even be implemented in other uses such as scanning the surface, in search and rescue missions and others. Lastly, with a flight computer on board, the use of AI can be implemented to limit the need for an operator.

9.3.4 Threats

There are two major challenges that the aircraft can face, specifically competition like quad-copters that do not require a runway to lift off. Any VTOL capable vehicle has an advantage over the airplane's lift-off capabilities because it is limited to the local infrastructure on the ground. Secondly, airspace is regulated through a country's laws and a special certification might have to be obtained for certain regions.

9.4 Financial viability

9.4.1 Risks

The business venture depends largely on the collaboration with disaster relief organizations, which are often non-profit organizations with government funding [34]. For this reason, risks include fluctuating funding, potential delays due to bureaucratic processes, and the vital need for a positive ethical perception. To avoid financial uncertainties, the aircraft must comply with all rules and laws in the given country of operation, diversify collaborations and establish contingency plans. The initiative must emphasize its positive impact on saving lives and the importance of disaster relief efforts.

Additionally, the development process must account for the fast-growing market, and be able to accompany new technology that may be developed in the future. Otherwise, the design will become outdated.

9.4.2 Cost of production

As mentioned thus far, the low cost of the aircraft is one of the main selling points. The de-

sign saves on production costs by excluding any composite materials, which are generally more expensive [35]. Furthermore, slats and an adjustable tail angle have been purposefully not included in the design to reduce the manufacturing costs of such complex components. To reduce the cost of material where unnecessary, the connection of the fuselage to the tail has been reduced to the size of two longerons, a smaller wing area has been opted for and rectangular wings that are easier to manufacture have been chosen.

9.4.3 Timeline - Scenario

The development phase of an aircraft this size is estimated to take between 5-7 years. Through this time, concrete definitions have to be set, a first test flight of the aircraft must be made and certification must be granted by a verified certifier before the airplane can go into service. Consider the example in Fig. 25.

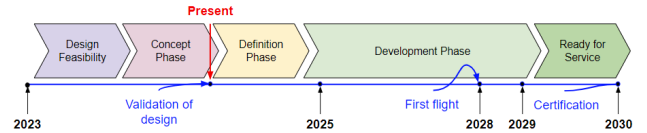


Figure 25: Timeline

Although, the prolonged development phase implies a more extended period of negative revenue, the expected market growth by 13.1% would result in Airbus being one of the leading companies in the emergency drone market.

9.4.4 Cost structure & benefit analysis

Airbus can make use of two main sources of revenue. Primarily the sale of the aircraft and a secondary source consisting of an optional service that offers regular replacement parts, in house repair and other features. A pay structure as such provides a great benefit in this situation, because the amount of disaster relief organizations that have access to a runway are limited [36]. Through this cost structure, the company is able to work with the customer, increasing the likelihood of retaining the customer as a client, while receive feedback and iteration an expansion into the global market. Taking into consideration the unfair advantage of the design, Airbus would reliably be able to save human lives and make a profit from the aircraft in a rapidly growing market.

10 Appendix B: Factsheet

Specification	Value	Specification	Value
Wing		Fuselage	
Airfoil type	NACA4412	Width	0.8 m
Wingspan	4.72 m	Height	0.6 m
Chord length	0.5 m	Length	1.14 m
Angle of attack	2°	Structural components material(stringers, frames, longerons)	Aluminium 7075
Wing weight	22 kg	Skin material	Aluminium 2024
L/D ratio	13.8	Ribs horizontal spacing	0.5 m
C_L during cruise with payload	0.7063	Fuselage weight	13.8 kg
Aspect ratio	7.84	Powertrain	
Structural components (spars, ribs, stringers) and upper skin material	Aluminium 7075	Motor max. power	20 kW
Lower skin material	Aluminium 2024	Motor max. RPM	6000 RPM
Enpenage		Motor diameter	0.2 m
Tail length (fuselage extension)	1 m	Motor horizontal distance from the tip of the nose	0.47 m
Tail rods length	1 m	Battery size	17.75 L
Horizontal and vertical stabiliser chord length	0.35 m	Battery density	150 Wh/kg
Horizontal and vertical stabiliser airfoil type	NACA0012	Battery weight	35.5 kg
Enpenage weight	9.2 kg	Motor weight	17 kg
Tail rods weight	6.6 kg	Flight electronics weight	4.8 kg
Tail rods outer diameter	8 cm	Propeller	
Tail rods inner diameter	7 cm	Airfoil	CLARK-Y
Landing gear		Type	GWS Slowfly 12" x 8"
Spacing from the ground	0.2 m	Diameter	0.85 m
Landing gear weight	8.1 kg	Number of blades	4
General		Blade angle offset	-1°
Center of gravity with and without the package (distance from the motor position)	0.5837 m	Propeller weight	1.4 kg
Total drag	41.1 N	Flight profile	
Total weight	143 kg	Cruise altitude	500 m
Payload weight	25 kg	Cruise velocity	150 km/h
Payload size	25 L	Loitering time	15 min

Table 5: Factsheet

11 Appendix C: Reflection

This reflection aims to provide a behind the scene preview of the team work and communication that has gone into creating this project. For some background, This project is from the minor in aircraft engineering, where students from different engineering disciplines are ordered into groups and tasked with designing an unmanned aircraft for disaster relief. This project group consists of 4 mechanical engineering students, 2 advanced technology students and 1 electrical engineering student.

The division of tasks has been made towards each student's strengths. Nick Potgieter and Maarten Post were in charge of calculating all the performance and stability parameters and flight profile due their great performance in the *Introduction to Aircraft Technology* assignments. Sven Onland was in charge of designing and calculating the fuselage structure. Bram Oudman was responsible for the design of the wing structure with an airfoil type given through the performance. Timo Janssen has done the power train design with the required power value from the performance calculations. Szczepan Letkiewicz has organized the business case and oversaw the project. Viktor Kulić Golja made the 3D CAD model of the designed airplane with the correct dimensions from the performance and structure calculations.

Overall, the group has come together and is collectively satisfied with the final outcome of the project. The reason for this opinion is that we have the basic concepts and dimensions for this plane for the specific task. A conceptual design that we have set out to create has been designed with the desired time frame, the results are more than satisfactory in our minds.

This does not go to say that the project has not had any challenges. A significant hurdle that the group faced was the sudden hospitalization of a team member. With less man power, the project went through a difficult phase, but the group was able to persevere. Aside from this hurdle, a problem that was encountered was working with many parameters that all depend on each other, making the calculations harder and leading to more loops. To try and overcome this issue, a common MATLAB online drive was created, but for the future a common variable naming system should have also been made to go along with it. To make an initial weight estimation was also hard, but afterwards the process went well. There were some challenges with the calculations of the structure and performance with the iterations after the original guess, but that has been solved with time.

With the project, all the group members have experienced how difficult it is to design an airplane which will fly, supported by calculations. Decision-making and assumptions were very important in this project based on engineering knowledge. Large amount of design iterations were made and had to be cut from the final report, since it aims to portray a conceptual design and not the design process. This made the last week of the project very dull to work on.

In the future, the experience of this project will be used in making educated estimations and finding ways to tackle problems. Also the self made scripts and functions will become useful in new design projects. The groups looks back on a hard worked project, but now knows a better way of planning and will be more efficient in another project.

12 Appendix D: Additional

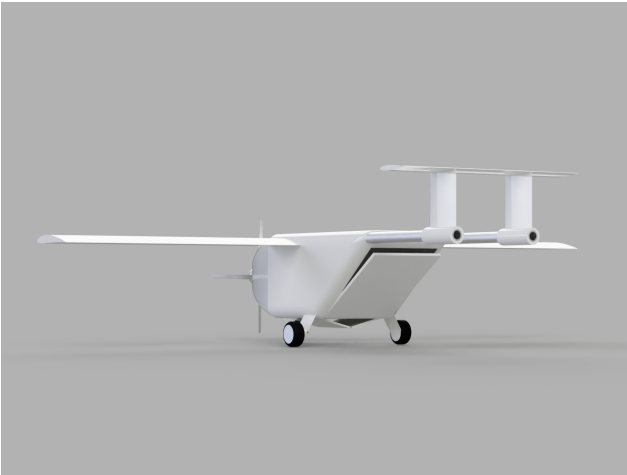


Figure 26: Render from the back

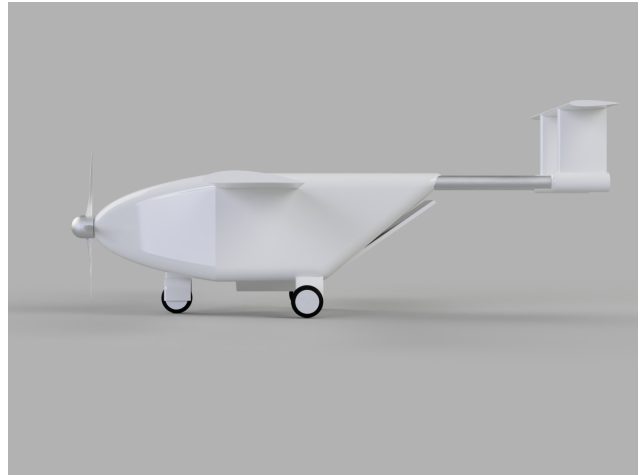


Figure 29: Render from the side

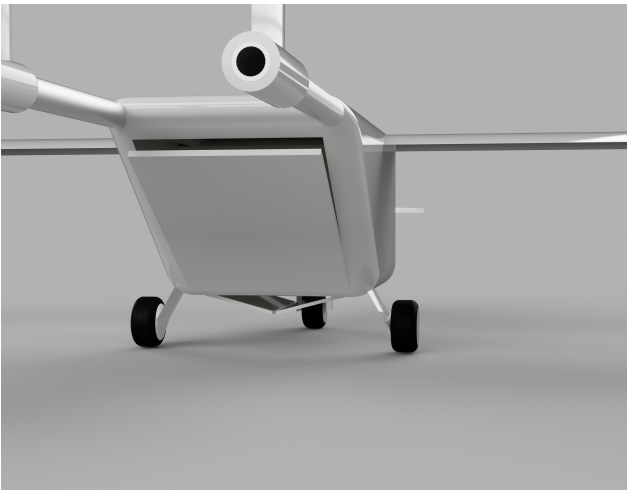


Figure 27: Render from the back

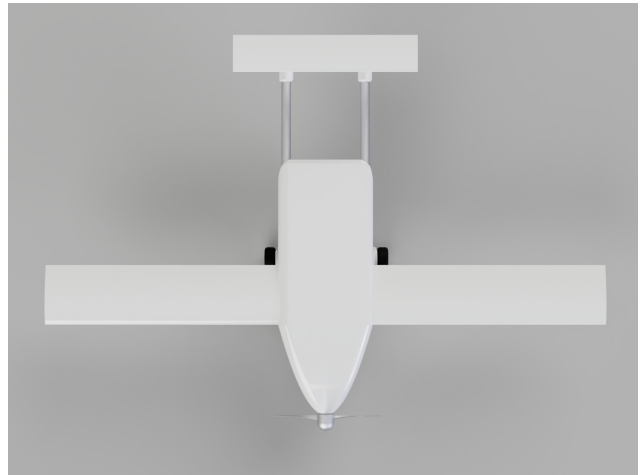


Figure 30: Render from the top

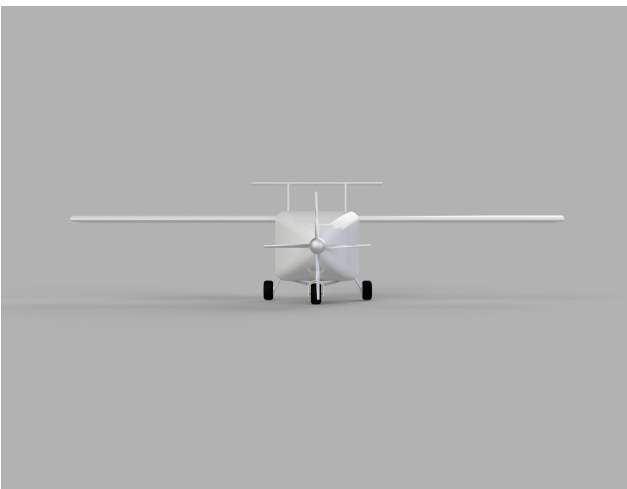


Figure 28: Render from the front

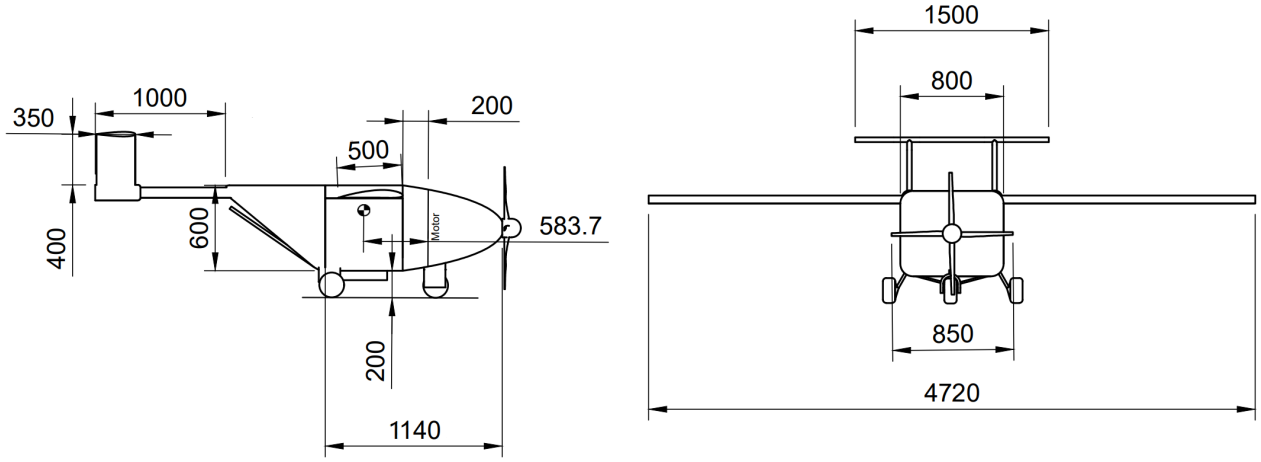


Figure 31: Technical drawing of the design with most important dimensions

12.1 Material properties

Table 6: Material properties Aluminum 2024. [16]

Parameter	Value
Yield strength	324 MPa
Young's Modulus	73.1 GPa
Poissons ratio	0.33
Density	2780 kg/m ³

Generative AI Disclosure

During the preparation of this work the authors used Grammarly [37] in order to improve grammar and check spelling. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the work.

Table 7: Material properties Aluminum 7075. [15]

Parameter	Value
Yield strength	71.7 GPa
Young's Modulus	455 MPa
Poissons ratio	0.33
Density	2810 kg/m ³

12.2 Dimensions

Table 8: Dimensions longerons inside fuselage

Parameter	Dimension [mm]
Top/bottom width	5
Web thickness	2
Top/bottom thickness	2
Height	7.5

Table 9: Dimensions stringers inside fuselage

Parameter	Dimension [mm]
Flange width	2
Thickness	1
Total height	2.5

12.3 Center of gravity location

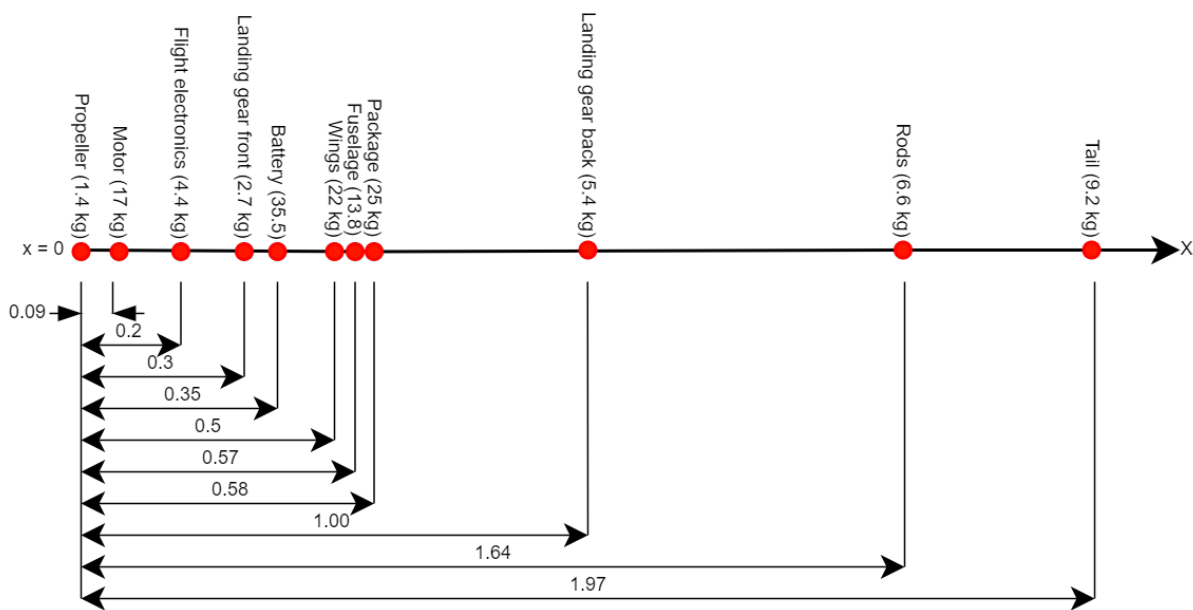


Figure 32: Center of gravity location

12.4 SWOT-Analys

SWOT - Analysis



Figure 33: SWOT-Analysis